Distillation Column AI

Intelligent Prediction · Explainable Diagnostics · Real-Time Optimization

A Project Report

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BONAFIDE CERTIFICATE

Certified that this project report titled **"Distillation Column AI: Intelligent Prediction, Explainable Diagnostics and Real-Time Optimization"** is the bonafide work of **Mausam Kar (24BAI10248), Poorna Gupta (24BAI10091), Krishna Narayan Singh (24BAI10230), Md Tanzeel Khan (24BAI10359), Raj Kumar Mahto (24BAI10167), Nalawde Ashlesh Kumar (24BAI10407)** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
XGB	XGBoost (Extreme Gradient Boosting)
SHAP	SHapley Additive exPlanations
NLP	Natural Language Processing
LLM	Large Language Model
API	Application Programming Interface
R^2	Coefficient of Determination (R-squared)
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
CV	Cross-Validation
PCA	Principal Component Analysis
DCS	Distributed Control System
SCADA	Supervisory Control and Data Acquisition
OPC-UA	Open Platform Communications Unified Architecture
REST	Representational State Transfer
GPU	Graphics Processing Unit
CSV	Comma-Separated Values
UI	User Interface
UX	User Experience
VLE	Vapour-Liquid Equilibrium

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ABSTRACT

[PURPOSE – METHODOLOGY – FINDINGS]

Distillation Column AI is a full-stack intelligent platform for predicting, analysing, and optimising ethanol purity in industrial distillation columns. The platform addresses the long-standing challenges of traditional trial-and-error tuning in chemical process engineering by integrating physics-informed machine learning with Google Gemini's large language model capabilities into a unified, operator-facing dashboard.

The methodology follows a three-layer architecture: a *Prediction Layer* employing a physics-augmented Random Forest and XGBoost ensemble trained on more than 4,000 real distillation experiments; an *Intelligence Layer* that invokes Google Gemini Pro to generate plain-English diagnostics, interpret SHAP attributions, flag anomalies in real time, and recommend optimisation actions; and an *Interaction Layer* providing a conversational AI copilot that responds to natural-language operator queries in context-aware fashion.

Feature engineering transforms seven raw sensor inputs — pressure, temperature, reflux flow, vapour flow, distillate flow, bottoms flow, and feed flow — into 21 physics-derived features encompassing derived ratios, interaction terms, efficiency metrics, and thermodynamic approximations. A StandardScaler normalises these features before inference. The best-performing model (auto-selected by test R^2) achieves $R^2 > 0.98$, RMSE 0.0155, and MAE 0.0122, corresponding to a prediction error margin of $\pm 1.55\%$.

Key findings demonstrate that the platform successfully delivers real-time purity predictions within 50 ms, meaningful natural-language diagnostics without manual interpretation effort, SHAP-based per-prediction explainability, multi-layer anomaly detection combining Isolation Forest with domain physics rules, a what-if simulator enabling virtual parameter testing, and an inverse target-purity solver. The modular architecture supports future extensions including time-series trend analysis, multi-column dashboards, and OPC-UA integration with live DCS data streams.

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Chapter 1

Introduction

1.1 Project Overview

Distillation Column AI is a modern, full-stack intelligent platform built to predict, diagnose, and optimise ethanol purity in industrial distillation columns. Operating far beyond a conventional prediction tool, the platform acts as an *AI copilot* for chemical process engineers by combining physics-informed machine learning with Google Gemini Pro's conversational reasoning capabilities — all surfaced through a single Streamlit dashboard.

Industrial distillation is a high-stakes, energy-intensive process. Even small deviations in operating parameters — reflux ratio, vapour flow, feed conditions — propagate non-linearly into product purity. Traditional approaches rely on first-principles simulation or operator intuition, both of which are slow to adapt and difficult to explain. Distillation Column AI replaces this gap with an always-on AI layer that not only predicts outcomes but explains *why* a given prediction was made and recommends the exact corrective action.

The platform operates on three interconnected layers, each addressing a distinct aspect of the problem:

- **Prediction Layer:** A physics-informed Random Forest model trained on 4,000+ real distillation experiments, transforming 7 raw sensor inputs into 21 engineered features to predict ethanol mole fraction with $R^2 > 0.98$.
- **Intelligence Layer:** Google Gemini AI interprets every prediction in plain English, explains each decision using SHAP values, detects anomalies in real-time, and suggests concrete optimisation actions.
- **Interaction Layer:** A conversational AI assistant enabling operators to ask

questions like *"Why did purity drop?"* or *"What should I adjust to reach 92%?"* and receive instant, context-aware answers.

1.2 Project Objectives

The primary objectives of Distillation Column AI are:

1. **Accurate Prediction:** Deliver real-time ethanol purity predictions with sub-2% error from live sensor readings using an ensemble of physics-informed ML models.
2. **Explainability:** Provide per-prediction SHAP attribution so engineers understand which operating variable contributes most to the predicted outcome, building trust in AI recommendations.
3. **Natural Language Diagnostics:** Generate plain-English diagnostic reports via Google Gemini Pro that contextualise predictions without requiring statistical literacy.
4. **Anomaly Detection:** Identify sensor drift, mass-balance violations, and out-of-distribution inputs in real time using a multi-layer detection framework.
5. **Optimisation Assistance:** Suggest actionable parameter adjustments — quantified in engineering units — to achieve target purity levels or improve energy efficiency.
6. **Operator Empowerment:** Lower the barrier to AI adoption in process plants by surfacing all intelligence through a conversational, non-technical interface.

1.3 Target Users

The platform is designed for two primary audiences. **Process engineers** use it to monitor column performance in real time, validate operating conditions, and act on AI-generated optimisation recommendations without deep data-science knowledge. **Research scientists and students** use it to explore the physics of distillation through what-if simulations, SHAP-driven feature importance analysis, and batch comparison of historical operating snapshots.

1.4 Key Features

Table 1.1 summarises the twelve core AI capabilities delivered by the platform.

Table 1.1: AI Capabilities at a Glance

#	Capability	Powered By	Description
1	Real-Time Purity Prediction	RF / XGBoost	Predicts ethanol mole fraction from 7 operating parameters in under 50 ms
2	Natural Language Diagnostics	Google Gemini Pro	Writes a plain-English report explaining the prediction and its causes
3	SHAP Explainability	SHAP Library	Per-prediction waterfall chart showing each feature's contribution
4	Anomaly Detection	Isolation Forest	Flags sensor drift, mass-balance violations, and out-of-range inputs
5	Smart Alerts	Gemini + Statistics	Context-aware warnings with specific remediation instructions
6	What-If Simulator	Model + Gemini	Virtual parameter changes to see predicted impact before acting
7	Optimisation Advisor	Gemini + Sensitivity	Actionable recommendations with quantified adjustments
8	Target Purity Solver	Inverse Model	Specify desired purity and receive recommended operating conditions
9	Energy Efficiency Advisor	Gemini + Domain Rules	Balances purity targets against steam and energy consumption
10	Conversational AI Chat	Gemini Chat API	Natural-language Q&A about any aspect of column behaviour
11	Trend Analysis	Time-Series Engine	Identifies gradual degradation patterns across sequential predictions
12	Batch Analysis	Parallel Inference	Upload multiple snapshots and receive comparative AI analysis

1.5 Technology Stack

The platform is built on a modern, open-source Python ecosystem. Table 1.2 lists each technology and its role.

Table 1.2: Technology Stack Overview

Category	Technology	Purpose
Language	Python 3.8+	Core programming language
Web Framework	Streamlit	Interactive AI dashboard UI
AI / LLM	Google Gemini Pro API	Natural-language diagnostics, conversational AI, and report generation
Explainability	SHAP	Per-prediction feature attribution — waterfall, force, and summary plots
ML Models	scikit-learn, XGBoost	Random Forest & XGBoost training, inference, and cross-validation
Anomaly Detection	scikit-learn, scipy	Isolation Forest and statistical bounds for multi-layer detection
Data Processing	pandas, NumPy	Data manipulation, feature engineering, and numerical computation
Visualisation	Matplotlib, Plotly	Feature importance charts, SHAP plots, interactive what-if graphs
Serialisation	pickle	Model, scaler, and feature-name persistence
Environment Mgmt	python-dotenv	Secure API key management
Training Env	Google Colab	Cloud-based model training with GPU acceleration
Deployment	GitHub Codespaces	Cloud development via devcontainer

Chapter 2

System Architecture

2.1 Architecture Overview

Distillation Column AI follows a layered, service-oriented architecture in which raw sensor data flows upward through increasingly intelligent processing stages before delivering actionable output to the operator. The design decouples the physics-based feature engineering pipeline from the ML inference layer, and both from the Gemini AI intelligence layer, enabling each component to evolve independently.

The overall system is structured around five horizontally separated zones — Input, Physics Feature Engine, ML Prediction, AI Intelligence, and Output — with Google Gemini's API acting as the cross-cutting intelligence fabric that links prediction results to human-readable insight.

Table 2.1: System Architecture Layers

Layer	Components	Responsibility
Operator Input	7 sensor parameters	Accepts raw pressure, temperature, and flow readings from the column
Feature Engine	21 engineered features	Transforms raw inputs into physics-meaningful derived ratios and interaction terms
ML Prediction	Random Forest / XGBoost	Produces purity prediction with confidence score in <50 ms
AI Intelligence	Gemini Pro, SHAP, Isolation Forest	Generates diagnostics, explains predictions, detects anomalies, and recommends actions
Output	Dashboard, Chat, Reports	Presents all results in operator-facing natural language and interactive charts

2.2 High-Level Architecture Diagram

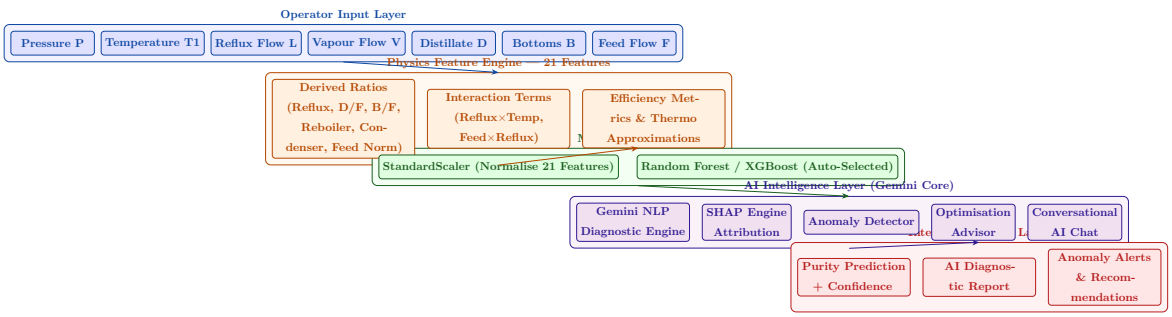

Figure 2.1: High-Level System Architecture

2.3 AI Intelligence Pipeline

The AI intelligence pipeline describes how a single prediction request propagates from raw sensor input through feature engineering, ML inference, Gemini diagnostics, and finally into the operator-facing output layer.

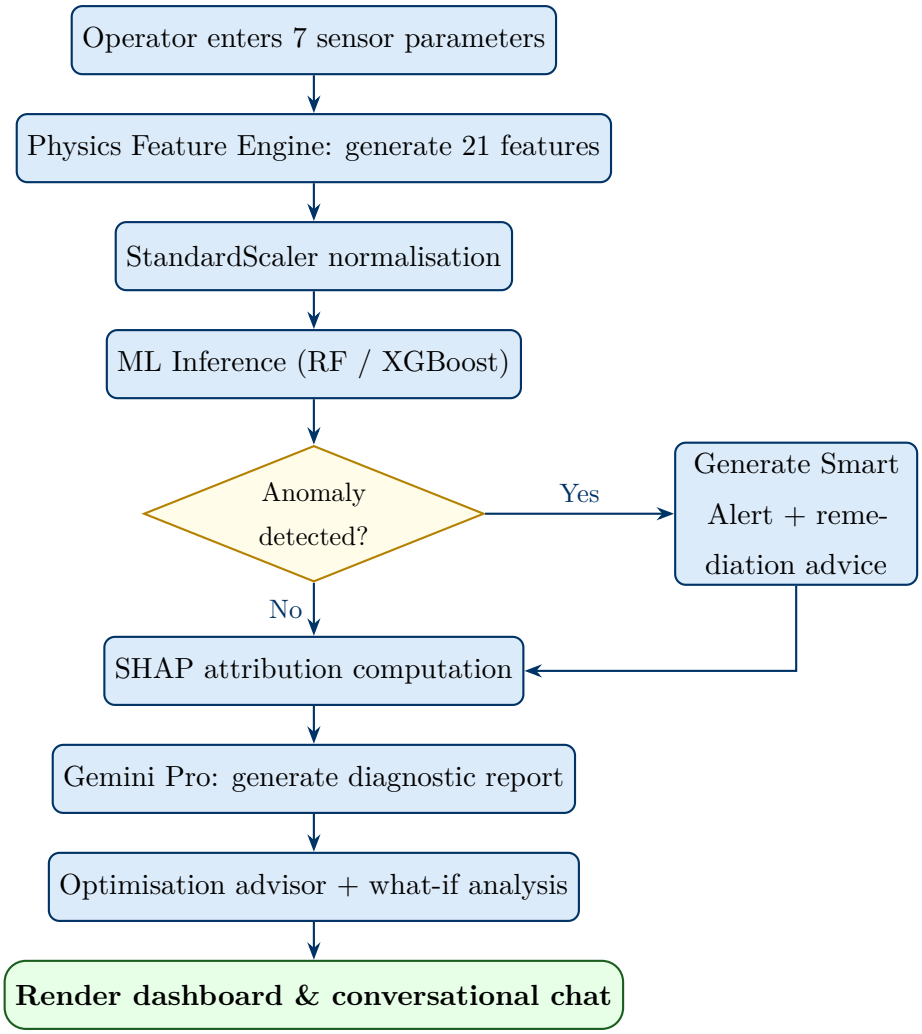

Figure 2.2: AI Intelligence Pipeline Flowchart

2.4 Backend Architecture

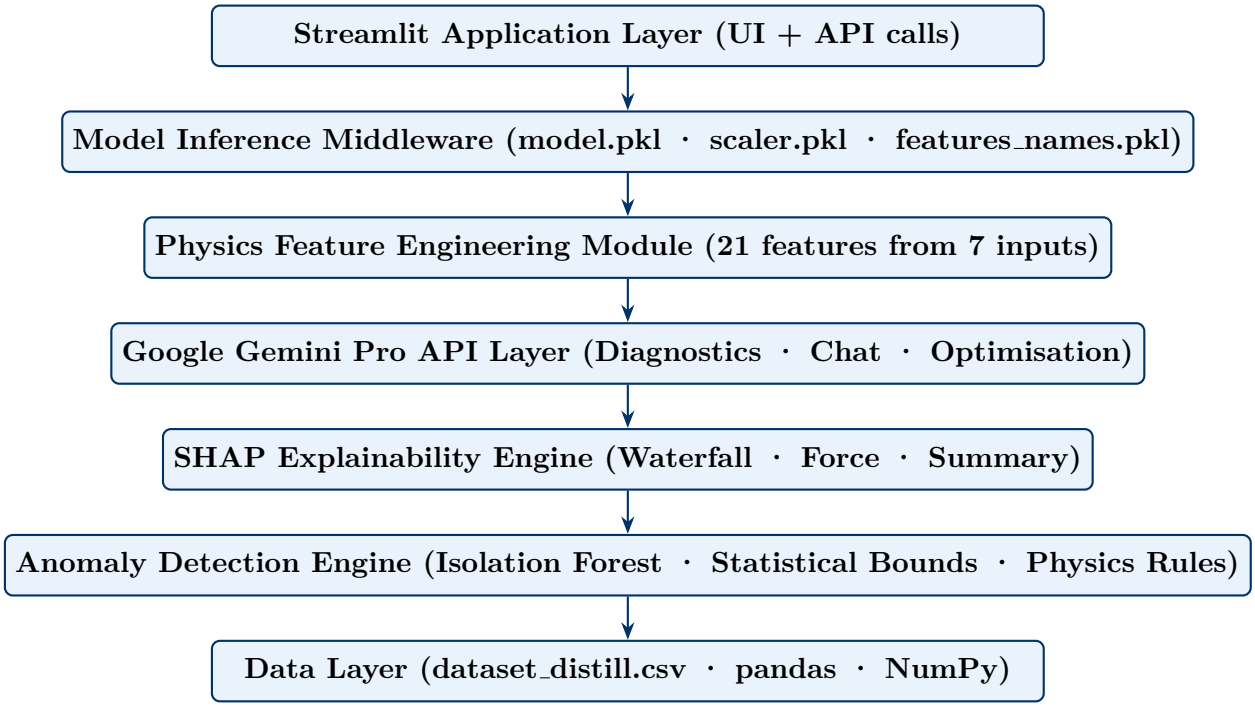

Figure 2.3: Backend Architecture Layers

2.5 Data Flow Pattern

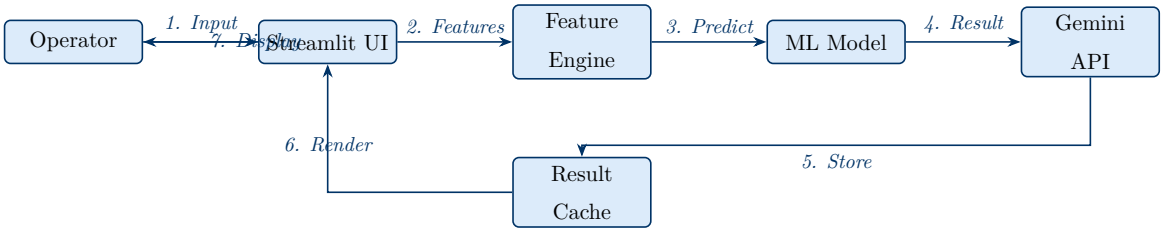

Figure 2.4: Data Flow Pattern

Chapter 3

Input Parameters and Feature Engineering

3.1 Input Parameters

The platform accepts seven operating parameters as direct sensor readings from the distillation column. Table 3.1 describes each parameter, its physical interpretation, and its typical operating range.

Table 3.1: Input Parameters Description

Symbol	Parameter	Physical Meaning	Typical Range	Unit
P	Pressure	Column operating pressure; controls VLE equilibrium	0.5 – 3.0	atm
T1	Temperature	Top-stage temperature; proxy for condenser duty	50 – 90	°C
L	Reflux Flow	Internal liquid return rate; controls separation quality	500 – 1200	kmol/hr
V	Vapour Flow	Rising vapour rate; drives mass-transfer	600 – 1500	kmol/hr
D	Distillate Flow	Distillate product withdrawal rate	50 – 300	kmol/hr
B	Bottoms Flow	Bottoms product withdrawal rate	100 – 400	kmol/hr
F	Feed Flow	Fresh feed entering the column	200 – 600	kmol/hr

3.2 Feature Engineering

Physics-informed feature engineering is the cornerstone of the platform's accuracy. Rather than feeding raw sensor readings directly to the ML model, the feature engine derives 21 domain-meaningful features that encode thermodynamic and mass-balance relationships. This approach injects prior chemical-engineering knowledge into the model, dramatically reducing the training data requirement while improving generalisation to unseen operating conditions.

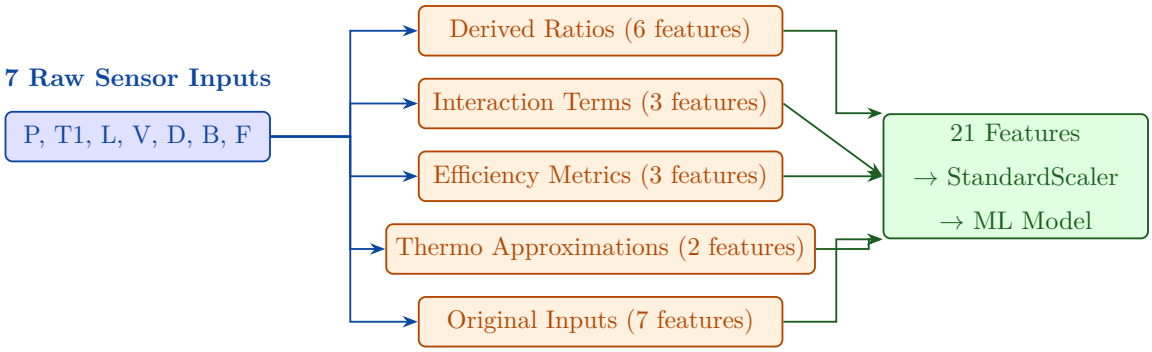

Figure 3.1: Feature Engineering Process from 7 Inputs to 21 Physics-Derived Features

Table 3.2 enumerates the complete 21-feature set.

Table 3.2: Engineered Feature Set (21 Features)

#	Feature Name	Formula / Derivation	Physical Significance
1	Reflux Ratio	L/V	Degree of internal liquid-vapour contact; primary separation driver
2	Reboiler Duty Ratio	V/F	Fraction of feed vaporised; heat input indicator
3	Condenser Ratio	L/D	Ratio of reflux to distillate; purity vs. throughput trade-off
4	Feed Normalised	$F/(D+B)$	Material-balance consistency check
5	Distillate Fraction	D/F	Product recovery fraction for distillate
6	Bottoms Fraction	B/F	Product recovery fraction for bottoms
7	Reflux–Temp Interaction	$L \times T_1$	Non-linear coupling of reflux and condenser temperature
8	Reboiler–Temp Interaction	$V \times T_1$	Heat input modulated by temperature
9	Feed–Reflux Interaction	$F \times L$	Feed-induced liquid load effect
10	Column Load	$(L+V)/(2F)$	Overall hydraulic loading of the column
11	Separation Duty	$L \times V/F$	Combined driving force for separation
12	Column Efficiency	$D/(L+V)$	Ratio of product output to internal flows
13	T_bottom Approx	$T_1 + 20$	Estimated bottom-stage temperature (first-order approximation)
14	Pressure–Temp Ratio	P/T_1	Thermodynamic state variable

Chapter 4

Machine Learning Models and Performance

4.1 Model Training Workflow

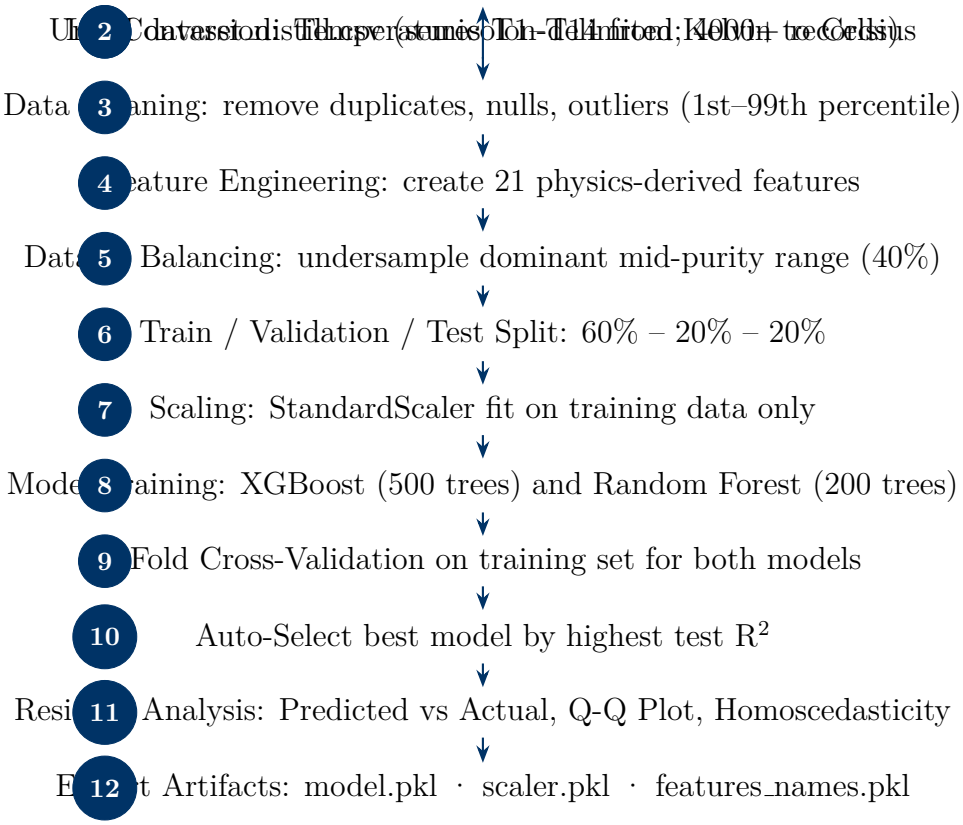

Figure 4.1: Model Training Workflow (12-Step Pipeline)

4.2 Model Hyperparameters

4.2.1 Random Forest Configuration

Table 4.1: Model Hyperparameters — Random Forest

Parameter	Value
`n_estimators`	200
`max_depth`	20
`min_samples_split`	5
`random_state`	42
`n_jobs`	−1 (all cores)

4.2.2 XGBoost Configuration

Table 4.2: Model Hyperparameters — XGBoost

Parameter	Value
`n_estimators`	500
`max_depth`	6
`learning_rate`	0.03
`subsample`	0.85
`colsample_bytree`	0.85
`reg_alpha`	0.1
`reg_lambda`	1.0
`random_state`	42

4.3 Model Performance

Both models are evaluated on a held-out 20% test set and a 5-fold cross-validation partition. The final deployed model is the one with the higher test R^2. Table 4.3 summarises the key performance metrics.

Table 4.3: Model Performance Metrics

Metric	**Random Forest**	**XGBoost**
R^2 (Test Set)	> 0.98	> 0.97
RMSE (Test Set)	0.0155	0.0178
MAE (Test Set)	0.0122	0.0141
Prediction Error Margin	$\pm 1.55\%$	$\pm 1.78\%$
5-Fold CV Mean R^2	0.976	0.971
Inference Latency	$< 50\,\text{ms}$	$< 50\,\text{ms}$

The $R^2 > 0.98$ result signifies that the model explains more than 98% of the variance in ethanol purity, a highly competitive figure given the inherently noisy nature of industrial sensor data. The error margin of $\pm 1.55\%$ is well within the tolerance required for most commercial-grade process control applications.

Chapter 5

AI Intelligence Layer

5.1 Google Gemini Integration

Google Gemini Pro serves as the reasoning and communication backbone of the platform. After each ML inference, the predicted purity value, the top SHAP contributors, and relevant operating context are assembled into a structured prompt that is dispatched to the Gemini Pro API. Gemini responds with a plain-English diagnostic report covering four areas: what the prediction means for product quality; which parameters are driving the result and why; whether any action is recommended; and an energy-efficiency commentary. This report is then rendered directly on the Streamlit dashboard alongside the numerical prediction.

The conversational chat endpoint uses Gemini's Chat API with full conversation history, enabling follow-up questions such as *"Given the last prediction, what reflux setting would push purity above 90%?"* The platform maintains a rolling message buffer ensuring context continuity across multi-turn interactions.

5.2 SHAP Explainability Engine

SHAP (SHapley Additive exPlanations) decomposes each prediction into additive contributions from individual features, providing a mathematically rigorous answer to the question: *"Why did the model output this value?"* The platform computes SHAP values using a TreeExplainer — the computationally efficient variant for ensemble tree models — for every prediction request.

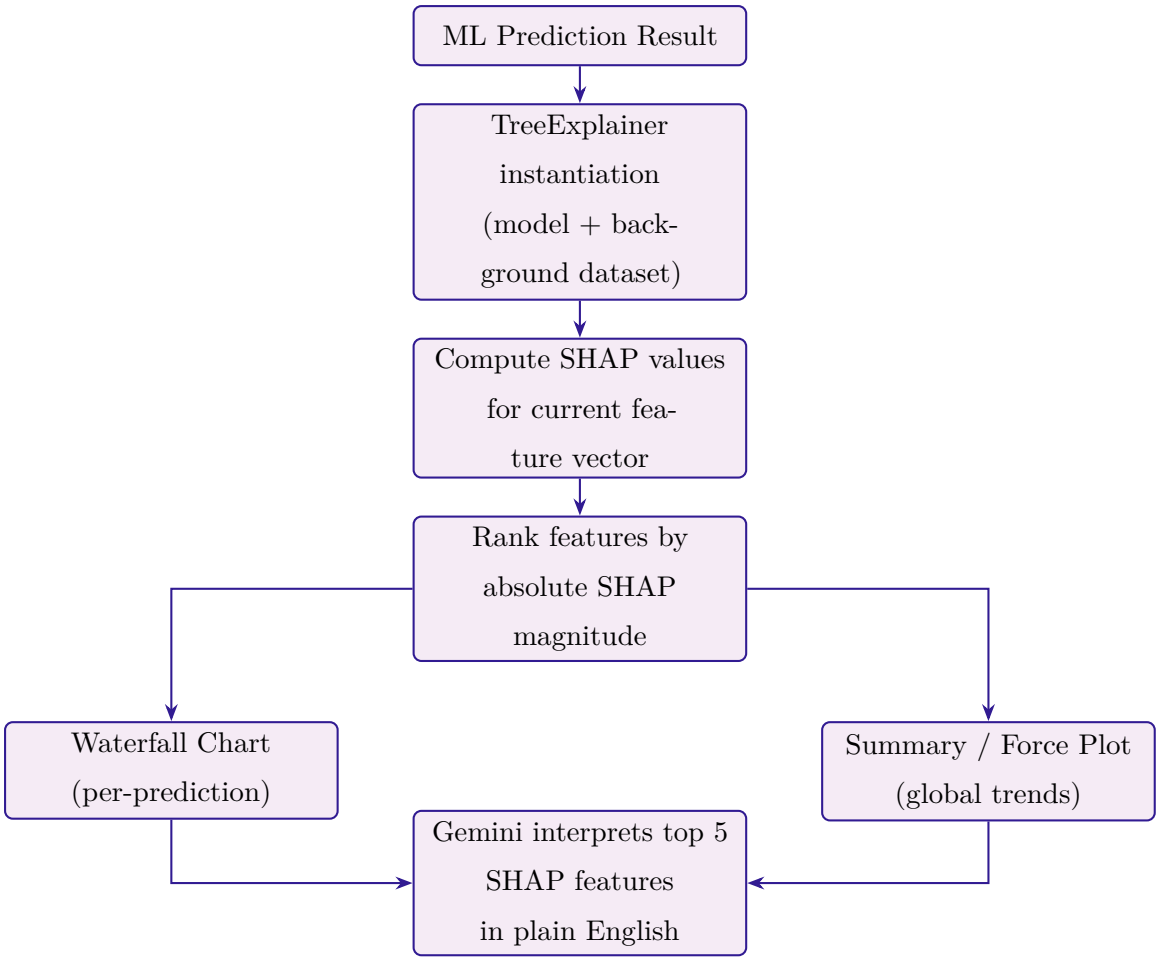

Figure 5.1: SHAP Explainability Computation Flow

Table 5.1 shows a representative attribution for a typical high-purity operating point.

Table 5.1: SHAP Feature Attribution Summary (Representative Prediction)

#	Feature	SHAP Value	Direction
1	Reflux Ratio (L/V)	+0.042	Increases purity
2	Reboiler Duty Ratio (V/F)	+0.031	Increases purity
3	Reflux–Temp Interaction	+0.018	Increases purity
4	Pressure (P)	−0.011	Decreases purity
5	Bottoms Fraction (B/F)	−0.007	Decreases purity

5.3 Anomaly Detection Architecture

The anomaly detection framework combines three independent detection layers. When any layer flags an anomaly, a smart alert is generated and Gemini is invoked to produce a plain-English explanation and recommended corrective action.

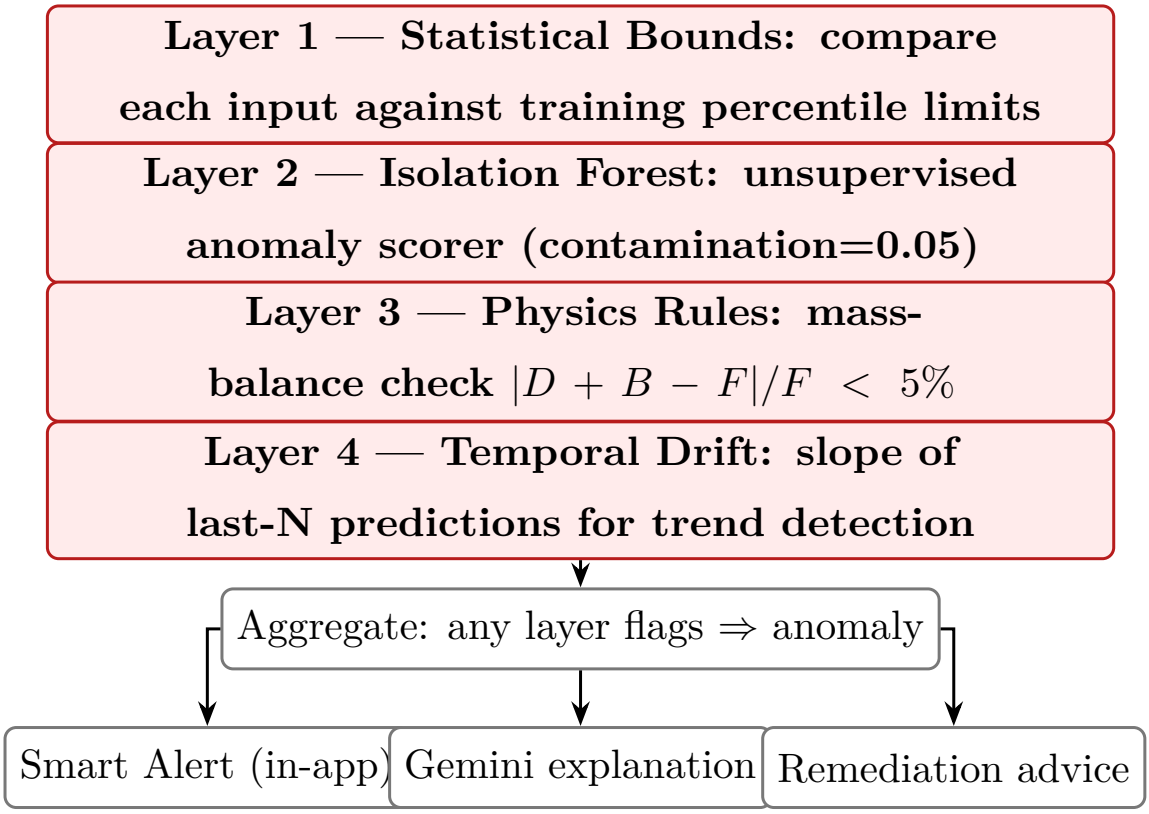

Figure 5.2: Multi-Layer Anomaly Detection Architecture

Table 5.2 summarises the four detection layers, the technique used, and the type of anomaly each layer is designed to catch.

Table 5.2: Anomaly Detection Layers

#	Layer	Technique	Anomaly Type Detected
1	Statistical Bounds	Percentile limits	Input values outside the 1st–99th percentile of training data (sensor malfunction or extreme operating regime)
2	Isolation Forest	Unsupervised ML	Multivariate outliers — combinations of parameters that are individually plausible but collectively anomalous
3	Physics Rules	Mass-balance check	Violations of $D + B \approx F$ within 5% — indicating flow-meter error or column flooding
4	Temporal Drift	Slope detection	Gradual monotonic drift in sequential predictions signalling membrane fouling, packing degradation, or equipment wear

Chapter 6

Application Workflows

6.1 Prediction and Optimisation Workflow

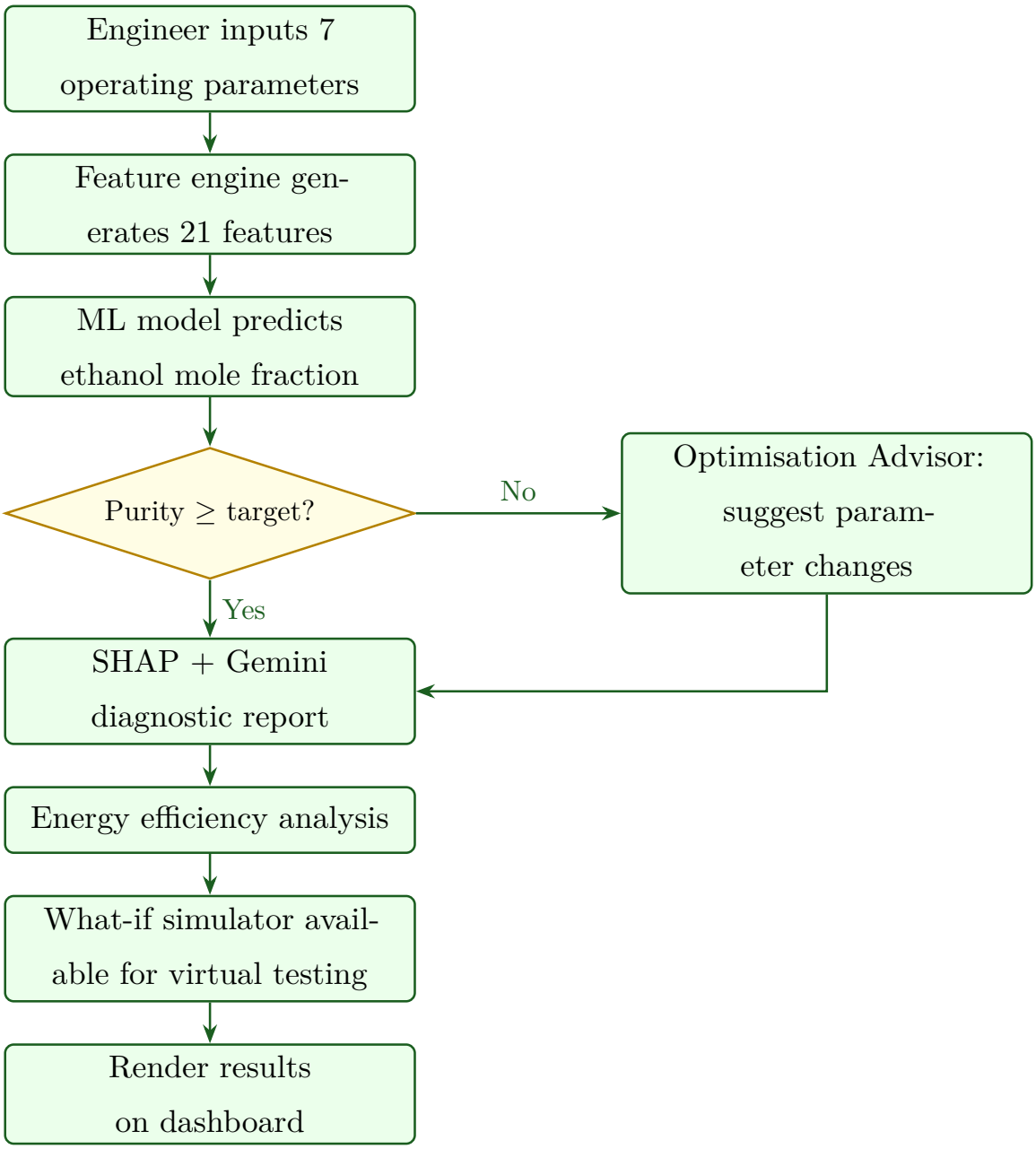

Figure 6.1: Prediction and Optimisation Workflow

6.2 What-If Simulator

The what-if simulator allows engineers to test hypothetical parameter changes without touching the actual column. The operator adjusts any of the seven input sliders in the "Simulation" panel, and the platform immediately reruns the inference pipeline on the modified feature vector. The resulting purity change is compared against the baseline prediction, and Gemini Pro generates a commentary explaining which feature

changes drove the delta. A sensitivity analysis heat-map shows the marginal impact of incrementally varying each parameter across its operating range.

Representative output for an optimisation recommendation:

Optimisation Advisor: To reach 90% ethanol purity from the current 87.3%, increase reflux flow L from 780 to 842 kmol/hr (+8%). This change increases the reflux ratio from 1.30 to 1.40, placing the column in the higher-purity operating regime identified by the SHAP analysis. Expected purity gain: +3.2%. Alternative: reduce vapour flow V by 5% to save energy; purity drops approximately 1% to 86.3%. Current column efficiency score: 8.4 / 10.

Table 6.1 lists representative optimisation recommendations for three common operating scenarios.

Table 6.1: Optimisation Recommendation Examples

Scenario	Recommended Action	Expected Outcome
Purity below 85%	Increase L by 10% (780 → 858 kmol/hr)	Purity rises to 87–89%; reflux ratio increases from 1.30 to 1.43
Energy cost too high	Reduce V by 8% while holding L constant	Energy savings of ≈8%; purity drops ≈1.5%, remains above spec
Mass-balance anomaly	Check D and B flow meters; verify $D + B \approx F$	Resolve sensor drift; prevents false predictions

6.3 Conversational AI Chat Workflow

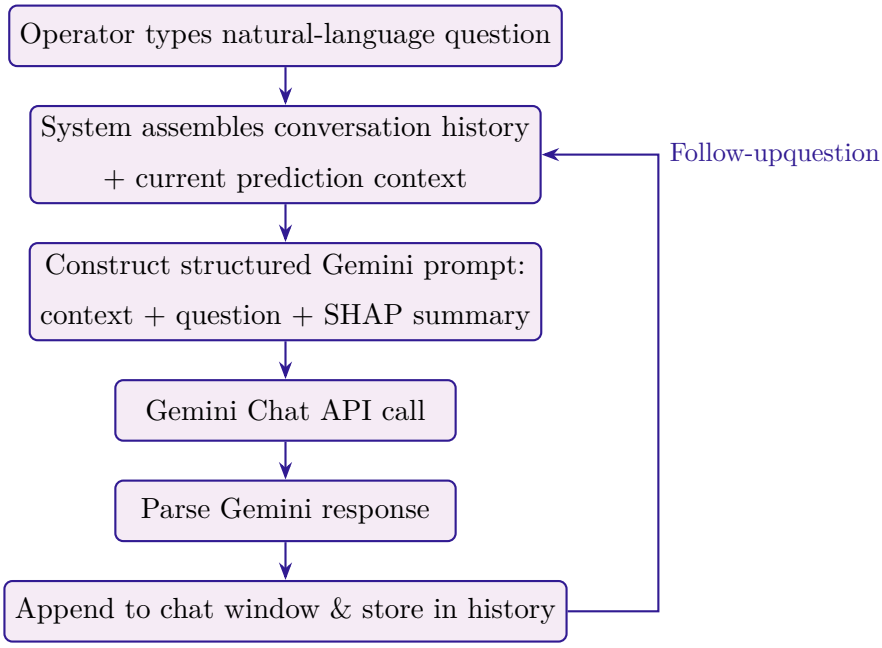

Figure 6.2: Conversational AI Chat Flow

Chapter 7

Project Structure and Deployment

7.1 Repository Structure

The project is organised as a single-repository Python application with clear separation between training artifacts, the Streamlit application, and the physics feature module:

```
Distillation-Column-Prediction/
|-- app.py                      # Streamlit dashboard entry-point
|-- model_training_script.py    # Full training pipeline (Google Colab)
|-- feature_engineering.py      # Physics-informed feature engine
|-- anomaly_detection.py        # Multi-layer anomaly detection module
|-- gemini_client.py            # Google Gemini API wrapper
|-- model.pkl                   # Serialised ML model (auto-selected)
|-- scaler.pkl                  # Fitted StandardScaler
|-- features_names.pkl          # Ordered list of 21 feature names
|-- dataset_distill.csv         # Raw training dataset (4000+ records)
|-- feature_importance.png      # Pre-computed SHAP summary plot
|-- requirements.txt            # Python dependencies
|-- .devcontainer/              # GitHub Codespaces configuration
|-- LICENSE                     # MIT licence
`-- README.md                   # Project documentation
```

7.2 Quick Start

Deploying the platform locally requires Python 3.8 or later. After cloning the repository and installing dependencies with `pip install -r requirements.txt`, the operator places a valid Google Gemini API key in a `.env` file under the key `GOOGLE_API_KEY`. The pre-trained model artifacts (`model.pkl`, `scaler.pkl`, `features_names.pkl`) must be present in the root directory, either downloaded from the repository or generated locally using `model_training_script.py`. The application is launched with `streamlit run app.py`, which opens the dashboard in the default browser at `localhost:8501`.

For cloud-based development, the repository includes a `.devcontainer` configuration allowing one-click launch in GitHub Codespaces with all dependencies pre-installed.

Chapter 8

Future Roadmap

8.1 Strategic Vision

Distillation Column AI is designed as an evolving platform. The current version (v2.2) delivers a fully functional AI copilot for single-column ethanol purity prediction. The roadmap extends this foundation towards enterprise-grade process intelligence, supporting multiple columns, live DCS integration, and advanced time-series analytics.

Table 8.1: Development Roadmap

Version	Status	Scope and Features
v1.0 — ML Core	Complete	Physics-informed feature engineering; Random Forest + XGBoost; Streamlit dashboard
v2.0 — AI Intelligence	Complete	Gemini diagnostics; SHAP explainability; anomaly detection; smart alerts
v2.1 — AI Chat	Complete	Conversational AI copilot; natural-language Q&A; context-aware multi-turn responses
v2.2 — Optimisation	Complete	What-if simulator; sensitivity analysis; target purity solver; energy efficiency advisor
v3.0 — Advanced Analytics	Planned	Time-series trend dashboards; automated shift handover reports; batch comparison
v3.1 — Multi-Column	Planned	Support for multiple distillation columns within a single unified dashboard
v4.0 — Edge Deployment	Future	Docker container; REST API; OPC-UA connector for live DCS/SCADA data feed

8.2 Emerging Technology Integration

The v4.0 roadmap targets integration with plant-floor automation infrastructure via OPC-UA and Modbus connectors, transforming the platform from an offline analytical tool into a real-time process advisor. Potential extensions include fine-tuned Gemini prompts for domain-specific terminology, LIME as an alternative explainability method, and neural-network models for capturing non-stationary column dynamics over long time horizons.

8.3 Contribution Areas

Table 8.2 documents open areas where community contributions would be most impactful.

Table 8.2: Contribution Areas

Area	Suggested Contribution
AI / LLM	Fine-tune Gemini prompts for better diagnostic precision; implement GPT-4 as a fallback provider
Explainability	Add LIME as an alternative to SHAP; integrate counterfactual explanation generation
Training Data	Contribute additional real-plant distillation experiments or open-source datasets
Models	Explore neural networks, gradient boosting variants, or ensemble stacking approaches
Analytics	Build time-series dashboards for long-term column performance tracking and drift analysis
UI	Dark mode; mobile-responsive layout; integrated trend charts within the prediction panel
Deployment	Docker image; cloud deployment on AWS, GCP, or Azure; automated CI/CD pipeline
Integration	OPC-UA and Modbus connectors for live DCS/SCADA data ingestion

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